

TYPES OF REACTIONS

Acids and Bases

Acids

- Contain the element H.
- Produce H^{1+} ions in solution.
- Have a sour taste.
- Turn blue litmus red.

e.g. HCl - hydrochloric
HNO₃ - nitric
H₂SO₄ - sulphuric
CH₃COOH - acetic
H₃PO₄ - phosphoric

Bases

- Any metal oxide or hydroxide.
- Produce OH^{1-} ions in solution.
- Have a soapy feel and bitter taste.
- Turn red litmus blue.

e.g. NaOH CaO
Mg(OH)₂ NH₄OH

Note

NH_4^{1+} behaves as if it is a metal!

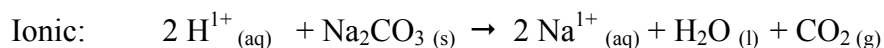
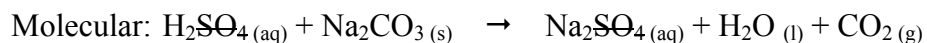
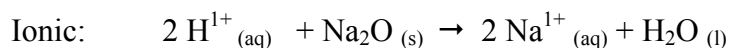
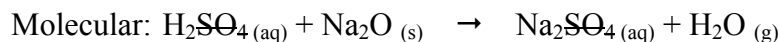
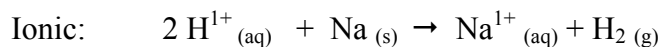
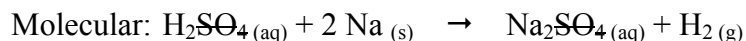
Acid Reactions

When an acid reacts with another substance, it produces a **salt**.

Salt: non-metal part of an acid combined with a metal or ammonium ion.

- HCl - produces **chloride** salts.
e.g. MgCl₂, NaCl, CuCl₂, NH₄Cl, etc.
- HNO₃ - produces **nitrate** salts.
e.g. Mg(NO₃)₂, NaNO₃, Cu(NO₃)₂, NH₄NO₃, etc.
- H₂SO₄ - produces **sulphate** salts.
e.g. MgSO₄, Na₂SO₄, CuSO₄, (NH₄)₂SO₄, etc.
- CH₃COOH - produces **acetate** salts.
e.g. Mg(CH₃COO)₂, NaCH₃COO, etc.

General Equations



Precipitation Reactions

When two soluble ionic solutions are mixed it is possible that one of the new combinations of ions produces an ***insoluble salt***. These types of reactions are known as ***precipitation reactions***.

To predict whether a precipitate will form, a table of ***solubility rules*** is used.

SOLUBLE

All Group I cations, NH_4^{1+}

All NO_3^{1-} , $\text{CH}_3\text{COO}^{1-}$ (acetate), ClO_3^{1-} (chlorates)

All Cl^{1-} , Br^{1-} , I^{1-} except Ag^{1+} , Pb^{2+} , Hg^{2+}

All SO_4^{2-} except Ba^{2+} , Sr^{2+} , Pb^{2+} , Ca^{2+} slightly soluble

INSOLUBLE

All O^{2-} , OH^{1-} except Group I, NH_4^{1+} , Ba^{2+} , Ca^{2+} , Sr^{2+} slightly soluble

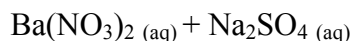
All S^{2-} except Group I, NH_4^{1+}

All CO_3^{2-} , PO_4^{3-} , SO_3^{2-} except Group I, NH_4^{1+}

Using The Solubility Rules

Barium nitrate and sodium sulphate solutions are mixed. Will a precipitate form?
If yes, write a molecular and net ionic equation to represent the reaction.

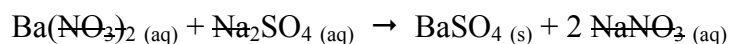
1. ***Write down the formula for each solution being mixed.*** They can both be indicated as (aq) because they are soluble:



2. ***Look at the two possible new combinations of ions, one at a time, and check out the solubility table to see if an insoluble substance is produced.***

The solubility rules indicate that, of the two possible products BaSO_4 and NaNO_3 , the BaSO_4 is insoluble.

3. ***Write out an equation that identifies the insoluble substance formed with an (s).*** The other possible combination can be represented with an (aq) to indicate that it remains in solution and is soluble.



4. ***Write out the net ionic equation by leaving out the spectator ions on both sides of the equation.***

